

PPVR: A Privacy-preserving Approach for User Behaviors in VR

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ABSTRACT

The increasing popularity of immersive environments creates a pressing challenge of reconciling service providers' desire to collect user behavior data with users' privacy concerns. In this study, we investigate the use of differential privacy (DP) algorithms in VR applications to enable statistical analysis of 3D spatial motion data while protecting against re-identification attacks. We assessed the efficacy of DP on both the original 3D spatial data and its cumulative heat map representation. This study underscores the utility of the DP algorithm in the context of 3D body motion data and highlights its potential for widespread adoption in diverse VR applications.

Index Terms: Human-centered computing; Security and privacy

1 INTRODUCTION

As VR devices become more widely adopted, researchers express increasing concerns about privacy issues in many Virtual Reality (VR) applications. On one hand, VR technology is rapidly evolving, and VR device companies understandably seek to continuously monitor user behavior data to improve user experiences. On the other hand, it can be argued that the privacy risks associated with VR applications are potentially more severe than those of mobile applications. This is due to the extensive collection of users' personal information by various input/output devices and sensors. Recent studies have revealed the ease with which attackers can identify and create profiles of VR users [4] using just a few minutes of data streaming. The immersive nature of VR contributes to these vulnerabilities, as it can make users more susceptible to self-disclosure and social engineering [3]. We argue that without a well-designed privacy protection technique, users' privacy is threatened by both client-side and server-side adversaries. In other words, these adversaries have the capability to access users' private data and easily re-identify them.

Differential privacy (DP) is a mathematical framework that provides a strong guarantee of privacy by allowing data to be analyzed without revealing sensitive information about any individual in the dataset [1, 5–7]. However, there is limited research on the use of DP in VR applications. To address this knowledge gap, we present two simple yet effective approaches to apply DP mechanism on 3D body motion data from VR applications. Our goal is to safeguard user privacy by preventing re-identification attack while retaining valuable statistical information, such as the body movements of users in a virtual scenario. Our first approach involves the direct application of DP onto the 3D body motion data while our second approach transforms the spatial data into 2D heat maps before applying DP. Our key contributions are as follows:

- We conduct experiments to apply Differential Privacy (DP) to 3D body motion data, aiming to protect user privacy while preserving data utility.
- We suggest a novel approach of transforming 3D body motion data into heat maps prior to applying DP, which enhances the efficacy of DP.

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- We assess our method using a real-world body motion dataset, specifically focusing on their resilience against re-identification attacks.

Our findings indicate that, despite the wealth of data being collected, user privacy can be preserved while still permitting data analysis for various purposes within an immersive environment. Given the rapid evolution of VR and sensor technologies, and the increasing collection of user behavior data for analysis, there is a pressing need for more research dedicated to privacy-preserving approaches, such as DP.

2 METHODOLOGY

In this section, we introduce our method of applying differential privacy onto 3D body motion data through (i) window slicing and (ii) converting the data into heat maps. Our approach aims to safeguard data privacy while preserving its utility.

2.1 Preparing Body Motion Data for DP

Outliers dropping and sequence alignment. Due to individual differences among participants and varying temporal lengths of 3D body motion data, we initially remove outliers and temporally align the sequences. Data falling outside three standard deviations are considered outliers and are removed from the dataset. We use the Dynamic Time Warping (DTW) method to align sequences, thereby preserving their distinct features.

Window slicing. Window slicing is a prevalent technique for training machine learning models on time series spatial data, operating on a rolling window with a predefined size, denoted as n , and a step size, denoted as m . This method iterates through each user's data recording, generating new samples at each step. We use a window size of $N = 10$ and the step size of $M = 1$ for our training and validation set for the evaluation. From an adversary's perspective, the application of the window slicing technique involves a majority vote among the labels of all sliced windows. This approach is frequently used as it enables the attack model to identify temporal patterns and dependencies within the data, and potentially minimize noise or inaccuracies in the prediction.

Heat map conversion. Alongside window slicing, we explore transforming the 3D body motion data into heat maps. This approach is frequently employed in the visualization of spatial data and consequently helps to preserve the utility of the data. Our proposed method comprises a series of steps. Initially, we calculate the range of data values on each axis, such as the X axis, represented by $[X_{max}, X_{min}]$. Subsequently, we define the resolution of the heat map as r , which determines the number of data points along each axis. By having r data points on each axis, the total number of data points in a heat map becomes $r \times r$. For each sample $x \in X$, we determine its position within the heat map using Equation 1. The "ceil" function, denoted as " $\lceil \cdot \rceil$ ", is applied to round each sample's position to the nearest integer, ensuring its placement within the heat map grid.

$$x_h = \lceil \frac{x - X_{min}}{X_{max} - X_{min}} * r \rceil \quad (1)$$

The value of each data point in the heat map is the number of samples that fall into the corresponding grid position. This provides a measure of the density or frequency of occurrences at that specific

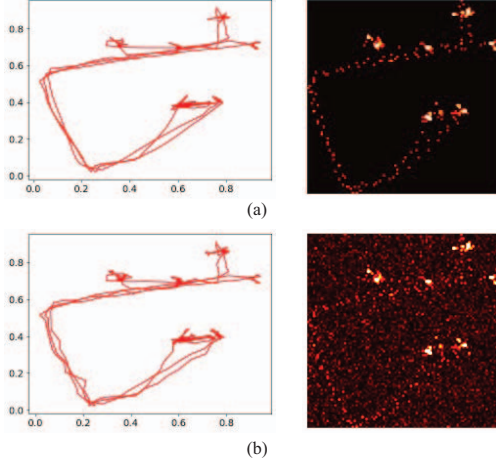


Figure 1: Qualitative evaluation of data utility (an example from VirtualHome dataset). (a) original data and corresponding heat map; (b) data visualization after applying Laplace differential privacy with $\epsilon = 3$.

location. Lastly, to ensure consistency and comparability, we normalize the data values in the heat map to fall within the range of $[0, 1]$. This step facilitates a standardized representation of the data across various heat maps. One disadvantage of converting body motion data into heat maps is the loss of temporal information, limiting the analysis to spatial movement tracing and frequency information alone. This is particularly true in scenarios such as virtual homes or virtual shopping, where the primary focus for application vendors may be recording movement traces and identifying the most frequently visited places by users.

2.2 Applying DP with Data Utility Preserved

We then apply differential privacy mechanism onto both the original motion data and the converted heat maps. To compare, we also conduct experiments applying DP on data with window slicing. In this study, we utilize the Diffprivlib implementation of the Laplace mechanisms, initially proposed by Dwork *et al.* [1]. We leverage these mechanisms to apply differential privacy to our data, ensuring privacy protection while retaining valuable information.

Utility preserving. Often referred to as the *privacy parameter* or *privacy budget*, ϵ represents the maximum permissible deviation between queries executed on two neighboring databases (d and d'), where $d, d' \in \mathcal{D}$ differ by only one data change (*i.e.*, the addition or removal of a single entry). More specifically, when ϵ is smaller, outputs generated for similar inputs must be very alike, thereby providing higher levels of privacy. On the other hand, when ϵ is larger, the outputs can exhibit greater dissimilarity, which leads to reduced privacy. Essentially, a smaller ϵ implies that more noise must be added to adequately protect the dataset's privacy. To balance between privacy protection and data utility, we introduce a data utility threshold to control the level of noise added through the application of differential privacy. This threshold helps us avoid excessive noise that could potentially compromise the usefulness of the data while still ensuring an acceptable level of privacy.

3 PRELIMINARY RESULTS

In this section, we present the experimental results of the data utility evaluation and the performance of our approach in protecting privacy against user identification attacks on VirtualHome dataset [2].

Data utility evaluation. We first evaluate the data utility through a qualitative method. This involves ensuring that the movement traces are still clear and recognizable to the human eyes after adding differential privacy noise. We demonstrate the data utility by plotting

Table 1: Privacy enhancement against user identification attack.

Methods	Models	User Identification Accuracy					
		Original	Privacy Enhanced with Differential Privacy				
			$\epsilon = 0.5$	$\epsilon = 1$	$\epsilon = 2$	$\epsilon = 5$	ϵ controlled by utility threshold
Window-slicing	LSTM	93.89%	27.15	36.51	53.07	75.91	48.03%
Heat map	CNN	96.53%	14.09	24.83	63.09	87.92	55.37%

the original spatial data in x/z space, along with its corresponding heat map. Specifically, to compare the data utility in different ϵ settings, we randomly sample from the dataset and plot its spatial data and corresponding heat maps, as shown in Figure 1(a). Then compare with data with differential privacy applied, as shown in Figure 1(b). From the experimental results, it is evident that higher privacy budgets introduce less noise into the data, which is expected as higher privacy budgets are applied. We further determine the data utility using RMSE thresholds. Specifically, we vary the values of ϵ . It can be observed that the RMSE decreases as ϵ increases. Moreover, there is an elbow point in the RMSE- ϵ curve, indicating that an appropriate RMSE threshold could be chosen near this point. **Privacy against user identification attacks.** We further evaluate the privacy protection performance of applying differential privacy to window-slicing data and its corresponding heat map. We conduct user identification attacks on the original data and privacy-enhanced data using various privacy budget settings, as well as an ϵ value selected based on the data utility threshold determined in the previous experiments. As shown in Table 1, the attack success rates experience a significant drop when differential privacy is applied. Specifically, the identification rate decreases in the range of from 41.16% to 45.86%, compared to the baseline attack success rates when no privacy enhancement is applied.

4 CONCLUSION AND FUTURE WORK

We present a research study focused on the application of DP to 3D body motion data within VR applications, with the goals of preserving both user privacy and data utility. The results indicate that our proposed heat map method surpasses the direct application of DP. These findings underscore the viability of our approach and advocate for further research into privacy-preserving techniques that do not compromise data utility in VR.

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